

Learning to Drive in a Day

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Abstract—We demonstrate the first application of deep reinforcement learning to autonomous driving. From randomly initialised parameters, our model is able to learn a policy for lane following in a handful of training episodes using a single monocular image as input. We provide a general and easy to obtain reward: the distance travelled by the vehicle without the safety driver taking control. We use a continuous, model-free deep reinforcement learning algorithm, with all exploration and optimisation performed on-vehicle. This demonstrates a new framework for autonomous driving which moves away from reliance on defined logical rules, mapping, and direct supervision. We discuss the challenges and opportunities to scale this approach to a broader range of autonomous driving tasks.

I. INTRODUCTION

Autonomous driving is a topic that has gathered a great deal of attention from both the research community and companies, due to its potential to radically change mobility and transport. Broadly, most approaches to date focus on formal logic which define driving behaviour in annotated 3D geometric maps. This can be difficult to scale, as it relies heavily on external mapping infrastructure rather than primarily using an understanding of the local scene.

In order to make autonomous driving a truly ubiquitous technology, we advocate for robotic systems which address the ability to drive and navigate in absence of maps and explicit rules, relying - just like humans - on a comprehensive understanding of the immediate environment [1] while following simple higher level directions (e.g., turn-by-turn route commands). Recent work in this area has demonstrated that this is possible on rural country roads, using GPS for coarse localisation and LIDAR to understand the local scene [2].

In the recent years, reinforcement learning (RL) – a machine learning subfield focused on solving Markov Decision Problems (MDP) [3] where an agent learns to select actions in an environment in an attempt to maximise some reward function – has shown an ability to achieve super-human results at games such as Go [4] or chess [5], a great deal of potential in simulated environments like computer games [6], and on simple tasks with robotic manipulators [7]. We argue that the generality of reinforcement learning makes it a useful framework to apply to autonomous driving. Most importantly, it provides a corrective mechanism to improve learned autonomous driving behaviour.

To this end, in this paper we:

- 1) pose autonomous driving as an MDP, explain how to design the various elements of this problem to

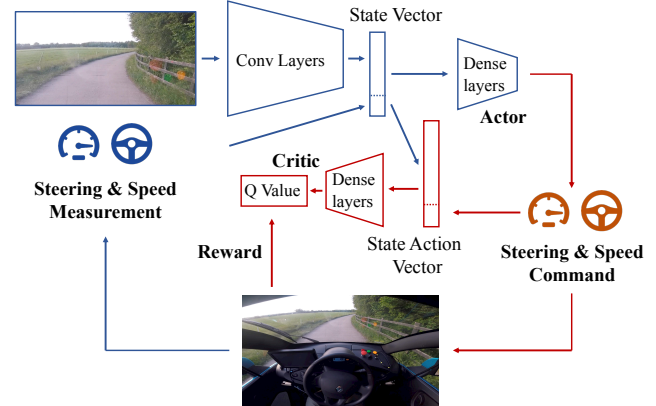


Fig. 1: We design a deep reinforcement learning algorithm for autonomous driving. This figure illustrates the actor-critic algorithm which we use to learn a policy and value function for driving. Our agent maximises the reward of distance travelled before intervention by a safety driver. A video of our vehicle learning to drive is available at <https://wayve.ai/blog/l2diad>

make it simpler to solve, whilst keeping it general and extensible,

- 2) show that a canonical RL algorithm (deep deterministic policy gradients [8]) can rapidly learn a simple autonomous driving task in a simulation environment,
- 3) discuss the system set-up required to make learning to drive efficient and safe on a real-world vehicle,
- 4) learn to drive a real-world autonomous vehicle in a few episodes with a continuous deep reinforcement learning algorithm, using only on-board computation.

We therefore present the first demonstration of a deep reinforcement learning agent driving a real car.

II. RELATED WORK

We believe this is the first work to show that deep reinforcement learning is a viable approach to autonomous driving. We are motivated by its potential to scale beyond that of imitation learning, and hope the research community examines autonomous driving from a reinforcement learning perspective more closely. The closest work in the current literature can predominantly be categorised as either imitation learning or classical approaches relying on mapping.

a) Mapping approaches: Since early examples [9], [10], autonomous vehicle systems have been designed to

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Abstract— 我们展示了深度强化学习在自动驾驶中的首次应用。从随机初始化的参数中，我们的模型能够使用单个单目图像作为输入在少数训练集中学习车道跟随策略。我们提供一个通用且容易获得的奖励：车辆在没有安全驾驶员控制的情况下行驶的距离。我们使用连续的、无模型的深度强化学习算法，所有探索和优化都在车辆上进行。这展示了自动驾驶的新框架，它摆脱了对定义的逻辑规则、映射和直接监督的依赖。我们讨论了将这种方法扩展到更广泛的自动驾驶任务中的挑战和机遇。

一、简介

自动驾驶因其有可能从根本上改变出行和交通而成为研究界和公司广泛关注的话题。从广义上讲，迄今为止大多数方法都侧重于形式逻辑，它定义带注释的 3D 几何地图中的驾驶行为。这可能很难扩展，因为它严重依赖外部地图基础设施，而不是主要使用对本地场景的理解。

为了使自动驾驶成为真正无处不在的技术，我们提倡机器人系统能够在没有地图和明确规则的情况下实现驾驶和导航的能力，就像人类一样，依赖于对周围环境的全面理解[1]，同时遵循简单的更高级别的指示（例如，逐向路线命令）。该领域最近的工作表明，这在乡村道路上是可能的，使用 GPS 进行粗略定位，并使用 LIDAR 来了解当地场景 [2]。

近年来，强化学习（RL）——一个专注于解决马尔可夫决策问题（MDP）[3]的机器学习子领域，其中代理学习在环境中选择动作以试图最大化某些奖励函数——已经显示出在围棋[4]或国际象棋[5]等游戏中实现超人类结果的能力，在计算机游戏[6]等模拟环境以及机器人操纵器[7]的简单任务中具有巨大的潜力。我们认为强化学习的普遍性使其成为适用于自动驾驶的有用框架。最重要的是，它提供了一种纠正机制来改善学习的自动驾驶行为。

为此，在本文中我们：

- 1) 将自动驾驶视为MDP，解释如何设计该问题的各个要素

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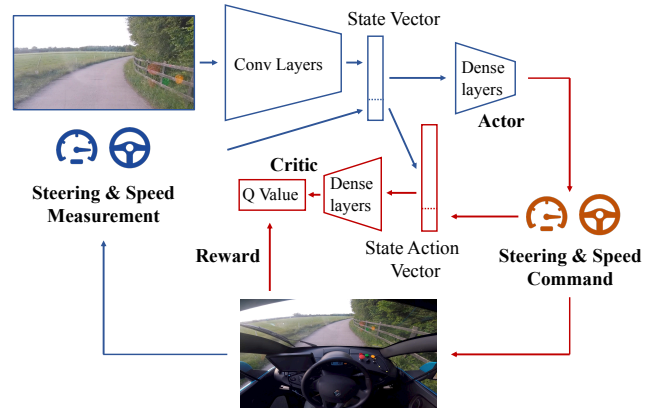


图1：我们设计了一种用于自动驾驶的深度强化学习算法。该图说明了我们用来学习驾驶策略和价值函数的演员批评算法。我们的代理将安全驾驶员干预之前行驶距离的奖励最大化。我们的车辆学习驾驶的视频可在 <https://wayve.ai/blog/l2diad> 上观看

- 使其更容易解决，同时保持其通用性和可扩展性，
- 2) 表明规范的强化学习算法（深度确定性策略梯度[8]）可以在模拟环境中快速学习简单的自动驾驶任务，
 - 3) 讨论在现实世界的车辆上学习驾驶高效和安全所需的系统设置，
 - 4) 仅使用车载计算，通过连续的深度强化学习算法在几个回合中学习驾驶现实世界的自动驾驶车辆。

因此，我们首次展示了驾驶真实汽车的深度强化学习代理。

二.相关工作

我们相信这是第一个证明深度强化学习是自动驾驶可行方法的研究。我们对其超越模仿学习的潜力感到鼓舞，并希望研究界从强化学习的角度更仔细地研究自动驾驶。当前文献中最接近的工作主要可以归类为模仿学习或依赖于映射的经典方法。

a) *Mapping approaches*: 自早期的示例 [9]、[10] 以来，自动驾驶车辆系统的设计目的是

navigate safely through complex environments using advanced sensing and control algorithms [11], [12], [13]. These systems are traditionally composed of many specific independently engineered components, such as perception, state estimation, mapping, planning and control [14]. However, because each component needs to be individually specified and tuned, this can be difficult to scale to more difficult driving scenarios due to complex interdependencies.

Significant effort has been focused on computer vision components for this modular approach. Localisation such as [15] facilitates control of the vehicle [16] within the mapped environment, while perception methods such as semantic segmentation [1] enable the robot to interpret the scene. These modular tasks are supported by benchmarks such as [17] and [18].

These modular mapping approaches are largely the focus of commercial efforts to develop autonomous driving systems; however, they present an incredibly complex systems engineering challenge, which has yet to be solved.

b) Imitation learning: A more recent approach to some driving tasks is imitation learning [19], [20], which aims to learn a control policy by observing expert demonstrations. One important advantage of this approach is that it can use end-to-end deep learning, optimising all parameters of a model jointly with respect to an end goal thus reducing the effort of tuning of each component. However, imitation learning is also challenging to scale. It is impossible to obtain expert examples to imitate for every potential scenario an agent may encounter, and it is challenging to deal with distributions of demonstrated policies (e.g., driving in each lane).

c) Reinforcement learning: Reinforcement learning is a broad class of algorithms for solving Markov Decision Problems (MDPs) [21]. An MDP consists of:

- a set $\mathcal{S}$ of states,
- a set $\mathcal{A}$ of actions,
- a transition probability function $p: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$, which to every pair $(s, a) \in \mathcal{S} \times \mathcal{A}$ assigns a probability distribution $p(\cdot|s, a)$ representing the probability of entering a state from state s using action a ,
- a reward function $R: \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, which describes the reward $R(s_{t+1}, s_t, a_t)$ associated with entering state s_{t+1} from state s_t using action a_t ,
- a future discount factor $\gamma \in [0, 1]$ representing how much we care about future rewards.

The solution of an MDP is a policy $\pi: \mathcal{S} \rightarrow \mathcal{A}$ that for every $s_0 \in \mathcal{S}$ maximises:

$$V_\pi(s_0) = \mathbb{E} \left(\sum_{t=0}^{\infty} \gamma^t R(s_{t+1}, s_t, \pi(s_t)) \right), \quad (1)$$

where the expectation is taken over states s_{t+1} sampled according to $p(s_{t+1}|s_t, \pi(s_t))$.

In our setting, we use a finite time horizon T in place of infinity in the above formula. This is equivalent to one of the states being terminal, i.e. it cannot be escaped and any action at that state gives zero reward.

Rearranging the above equation into a recurrent form, we get one of the two Bellman equations:

$$V_\pi(s_0) = \mathbb{E} \left(R(s_1, s_0, \pi(s_0)) + \gamma V_\pi(s_1) \right). \quad (2)$$

Here the expectation is taken only over s_1 sampled according to $p(s_1|s_0, \pi(s_0))$. For reference, let us present the other Bellman equation:

$$Q_\pi(s_0, a_0) = \mathbb{E} \left(R(s_1, s_0, a_0) + \gamma Q_\pi(s_1, \pi(s_1)) \right), \quad (3)$$

where $Q_\pi(s_0, a_0)$ is the expected cumulative discounted reward received while starting from state s_0 with action a_0 and following policy π thereafter. Again the expectation is taken over s_1 sampled according to $p(s_1|s_0, a_0)$.

In other words, reinforcement learning algorithms aim to learn a policy π that obtains a high cumulative reward. They are generally split into two categories: model-based and model-free reinforcement learning. In the former approach, explicit models for the transition and reward functions are learnt, and then used to find a policy that maximises cumulative reward under those estimated functions. In the latter, we directly estimate the value $Q(s, a)$ of taking action a in state s , and then follow a policy that selects the action with the highest estimated value in each state.

Model-free reinforcement learning is extremely general. Using it, we can (in theory) learn any task we can imagine, whereas model-based algorithms can be only as good as the model learned. On the other hand, model-based methods tend to be more data-efficient than model-free ones. For further discussion, see [22].

In autonomous driving, deep learning has been used to learn dynamics models for model-based reinforcement learning using off-line data [23]. Reinforcement learning has also been used to learn autonomous driving agents in video games. However, this can simply the problem, with access to ground truth reward signals which are not available in the real-world, such as the angle of the car to the the lane [8].

The closest work to this paper is from Riedmiller et al. [24] who train a reinforcement learning agent which drives a vehicle to follow a GPS trajectory in an obstacle-free environment. They demonstrate learning on-board the vehicle using a dense reward function based on GPS thresholded tracking error. We build on this work in a number of ways; we demonstrate learning to drive with deep learning, from an image-based input, using a sparse reward function to lane follow.

III. SYSTEM ARCHITECTURE

A. Driving as a Markov Decision Process

A key focus of this paper is the set-up of driving as an MDP. Our goal is that of autonomous driving, and the exact definition of the state space $\mathcal{S}$, action space $\mathcal{A}$ and reward function R are free for us to be defined. The transition model is implicitly fixed once a state and action representation is fixed, with the remaining degrees of freedom – the

使用先进的传感和控制算法在复杂的环境中安全导航[11]、[12]、[13]。这些系统传统上由许多特定的独立设计的组件组成，例如感知、状态估计、绘图、规划和控制[14]。然而，由于每个组件都需要单独指定和调整，因此由于复杂的相互依赖性，这可能很难扩展到更困难的驾驶场景。

这种模块化方法的计算机视觉组件已投入大量精力。[15]等定位有助于在地图环境中控制车辆[16]，而语义分割[1]等感知方法使机器人能够解释场景。这些模块化任务得到[17]和[18]等基准测试的支持。

这些模块化地图方法在很大程度上是开发自动驾驶系统的商业努力的重点。然而，它们提出了极其复杂的系统工程挑战，尚未解决。

b) Imitation learning: 一些驾驶任务的最新方法是模仿学习[19]、[20]，其目的是通过观察专家演示来学习控制策略。这种方法的一个重要优点是它可以使用端到端深度学习，针对最终目标联合优化模型的所有参数，从而减少调整每个组件的工作量。然而，模仿学习也难以规模化。不可能获得专家示例来模仿代理可能遇到的每种潜在场景，并且处理演示策略的分布（例如，在每个车道上行驶）具有挑战性。

c) Reinforcement learning: 强化学习是一类用于解决马尔可夫决策问题（MDP）的广泛算法[21]。MDP 包括：

- 状态集 $\mathcal{S}$,
- 一组 $\mathcal{A}$ 动作,
- 转移概率函数 $p: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$ ，它为每对 $(s, a) \in \mathcal{S} \times \mathcal{A}$ 分配一个概率分布 $p(\cdot|s, a)$ ，表示使用动作 a 从状态 s 进入状态的概率，
- 奖励函数 $R: \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ ，它描述与使用操作 a_t 从状态 s_t 进入状态 s_{t+1} 相关的奖励 $R(s_{t+1}, s_t, a_t)$ ，
- 未来折扣因子 $\gamma \in [0, 1]$ 代表我们对未来奖励的关心程度。

MDP 的解决方案是策略 $\pi: \mathcal{S} \rightarrow \mathcal{A}$ ，对于每个 $s_0 \in \mathcal{S}$ 最大化：

$$V_\pi(s_0) = \mathbb{E} \left(\sum_{t=0}^{\infty} \gamma^t R(s_{t+1}, s_t, \pi(s_t)) \right), \quad (1)$$

其中期望取自根据 $p(s_{t+1}|s_t, \pi(s_t))$ 采样的状态 s_{t+1} 。

在我们的设置中，我们使用有限时间范围 T 代替上述公式中的无穷大。这相当于其中一个状态是终结状态，即它无法逃脱，并且该状态下的任何操作都会给出零奖励。

将上述方程重新整理成递归形式，我们得到两个贝尔曼方程之一：

$$V_\pi(s_0) = \mathbb{E} \left(R(s_1, s_0, \pi(s_0)) + \gamma V_\pi(s_1) \right). \quad (2)$$

这里的期望仅在根据 $p(s_1|s_0, \pi(s_0))$ 采样的 s_1 上进行。作为参考，让我们提出另一个贝尔曼方程：

$$Q_\pi(s_0, a_0) = \mathbb{E} \left(R(s_1, s_0, a_0) + \gamma Q_\pi(s_1, \pi(s_1)) \right), \quad (3)$$

其中 $Q_\pi(s_0, a_0)$ 是从状态 s_0 开始执行操作 a_0 并随后遵循策略 π 时收到的预期累积折扣奖励。再次根据 $p(s_1|s_0, a_0)$ 采样 s_1 来获取期望

换句话说，强化学习算法的目标是学习获得高累积奖励的策略 π 。它们通常分为两类：基于模型的强化学习和无模型的强化学习。在前一种方法中，学习了转换和奖励函数的显式模型，然后用于找到在这些估计函数下最大化累积奖励的策略。在后者中，我们直接估计在状态 s 下采取动作 a 的值 $Q(s, a)$ ，然后遵循在每个状态下选择具有最高估计值的动作的策略。

无模型强化学习非常普遍。使用它，我们（理论上）可以学习我们可以想象的任何任务，而基于模型的算法只能与所学习的模型一样好。另一方面，基于模型的方法往往比无模型的方法更有效。进一步的讨论请参见[22]。

在自动驾驶中，深度学习已被用来学习动态模型，以使用离线数据进行基于模型的强化学习[23]。强化学习也被用来学习视频游戏中的自动驾驶代理。然而，这可以简单地解决问题，因为可以访问现实世界中无法获得的地面真实奖励信号，例如汽车与车道的角度[8]。

与本文最接近的工作来自 Riedmiller 等人。[24] 他们训练了一个强化学习代理，该代理驱动车辆在无障碍环境中遵循 GPS 轨迹。他们使用基于 GPS 阈值跟踪误差的密集奖励函数演示了车载学习。我们通过多种方式以这项工作为基础；我们展示了通过深度学习学习驾驶，从基于图像的输入，使用稀疏奖励函数进行车道跟随。

三. 系统架构

A. Driving as a Markov Decision Process

本文的重点是将驾驶设置为 MDP。我们的目标是自动驾驶，状态空间 $\mathcal{S}$ 、动作空间 $\mathcal{A}$ 和奖励函数 R 的准确定义是我们可以自由定义的。一旦状态和动作表示固定，转换模型就隐式固定，其余自由度 -

transitions themselves – dictated by the mechanics of the simulator/vehicle used.

a) State space: Key to defining the state space is the definition of the observations O_t that the algorithm receives at each time step. Many sensors have been developed in order to provide sophisticated observations for driving algorithms, not limited to LIDAR, IMUs, GPS units and IR depth sensors; an endless budget could be spent on advanced sensing technology. In this paper, we show that for simple driving tasks it is sufficient to use a monocular camera image, together with the observed vehicle speed and steering angle. Theoretically, state s_t is to be a Markov representation of all previous observations. An approximation a fixed length approximately Markov state could be obtained by, for example, using a Recurrent Neural Network to recursively combine observations. However, for the tasks we consider, the observation itself serves as a good enough approximation of the state.

A second consideration is how to treat the image itself: the raw image could be fed directly into the reinforcement learning algorithm through a series of convolutions [25]; alternatively, a small compressed representation of the image, using, for example, a Variational Autoencoder (VAE) [26] [27], could be used. We compare the performance of reinforcement learning using these two approaches in Section IV. In our experiments, we train the VAE online from five purely random exploration episodes, using a KL loss and a L2 reconstruction loss [27].

b) Action space: Driving itself has what one might think are a natural set of actions: throttle, brake, signals etc. But what domain should the output of the reinforcement learning algorithm be? The throttle itself can be described as discrete, either on or off, or continuous, in a range isometric to $[0, 1]$. An alternative is to reparameterise the throttle in terms of a speed set-point, with throttle output by a classical controller in an attempt to match the set-point. Overall, experiments on a simple simulator (Section IV-A) showed that continuous actions, whilst somewhat harder to learn, provide for a smoother controller. We use a two-dimensional action space; steering angle in the range $[-1, 1]$ and speed setpoint in km/h.

c) Reward function: Design of reward functions can approach supervised learning – given a lane classification system, a reward to learn lane-following can be set up in terms of minimising the predicted distance from centre of lane, the approach taken in [8]. This approach is limited in scale: the system can only be as good as the human intuition behind the hand-crafted reward. We do not take this approach. Instead, we define the reward as forward speed and terminate an episode upon an infraction of traffic rules – thus the value of a given state $V(s_t)$ corresponds to the average distance travelled before an infraction. A fault that may be identified is that the agent may choose to avoid more difficult manoeuvres, e.g. turning right in the UK (left in US). Command conditional rewards may be utilised in future work to avoid this.

B. Reinforcement Learning Algorithm – Deep Deterministic Policy Gradients

We selected a simple continuous action domain model-free reinforcement learning algorithm: deep deterministic policy gradients (DDPG) [8], to show that an off-the-shelf reinforcement learning algorithm with no task-specific adaptation is capable of solving the MDP posed in Section III-A.

DDPG consists of two function approximators: a critic $Q: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, which estimates the value $Q(s, a)$ of the expected cumulative discounted reward upon using action a in state s , trained to satisfy the Bellman equation

$$Q(s_t, a_t) = r_{t+1} + \gamma(1 - d_t)Q(s_{t+1}, \pi(s_{t+1})),$$

under a policy given by the actor $\pi: \mathcal{S} \rightarrow \mathcal{A}$, which attempts to estimate a Q -optimal policy $\pi(s) = \arg\max_a Q(s, a)$; here $(s_t, a_t, r_{t+1}, d_{t+1}, s_{t+1})$ is an experience tuple, a transition from state s_t to s_{t+1} using action a_t and receiving reward r_{t+1} and “done” flag d_{t+1} , selected from a buffer of past experiences. The error in the Bellman equality, which the critic attempts to minimise, is termed the temporal difference (TD) error. Many variants of actor-critic methods exist, see e.g. [28], [29].

DDPG training is done online. Beyond the infrastructure of setting up such a buffer for use on a real vehicle (which requires it to be tolerant of missing/faulty episodes and any-time stoppable), reinforcement learning can be sped up by selecting the most “informative” examples from the replay buffer. We do so using a commonly established method called prioritised experience replay [30]: we sample experience tuples with probability proportional to the TD error made by the critic. The weights used for this sampling are updated upon each optimisation step with minimal overhead; new samples are given infinite weight to ensure all samples are seen at least once.

DDPG is an off-policy learning algorithm, meaning that actions performed during training come from a policy distinct from the learn optimal policy by the actor. This happens in order to gain diverse state-action data outside of the narrow distribution that would be seen by the optimal policy, and thus increase robustness. We use a standard method of achieving this in the context of continuous reinforcement learning methods: our exploration policy is formed by adding discrete Ornstein-Uhlenbeck process noise [31] to the optimal policy. Therefore, at each step we add to optimal actions noise x_t given by:

$$x_{t+1} = x_t + \theta(\mu - x_t) + \sigma \epsilon_t, \quad (4)$$

where θ, μ, σ are hyperparameters and $\{\epsilon_t\}_t$ are i.i.d. random variables sampled from the normal distribution $N(0, 1)$. These parameters need to be tuned carefully, as there is a direct trade-off between noise utility and comfort of the safety driver. Strongly mean reverting noise with lower variance is easier to anticipate, whilst higher variance noise provides better state-action space coverage.

转换本身——由所使用的模拟器/车辆的机制决定。

a) *State space*: 定义状态空间的关键是算法在每个时间步接收的观测值 O_t 的定义。为了为驾驶算法提供复杂的观测，已经开发了许多传感器，不仅限于激光雷达、IMU、GPS 单元和红外深度传感器；先进的传感技术可以花费无穷无尽的预算。在本文中，我们表明，对于简单的驾驶任务，使用单目相机图像以及观察到的车速和转向角就足够了。理论上，状态 s_t 是所有先前观察结果的马尔可夫表示。例如，可以通过使用递归神经网络递归地组合观测值来获得近似马尔可夫状态的固定长度的近似值。然而，对于我们考虑的任务，观察本身就足以近似状态。

第二个考虑因素是如何处理图像本身：原始图像可以通过一系列卷积直接输入到强化学习算法中[25]；或者，可以使用图像的小型压缩表示，例如使用变分自动编码器（VAE）[26][27]。我们在第四节中比较了使用这两种方法的强化学习的性能。在我们的实验中，我们使用 KL 损失和 L2 重建损失，从五个纯随机探索事件中在线训练 VAE [27]。

b) *Action space*: 驾驶本身具有人们可能认为的一组自然动作：油门、刹车、信号灯等。但是强化学习算法的输出应该在哪个域？节流阀本身可以描述为离散的（打开或关闭），或者连续的（在 $[0, 1]$ 等距范围内）。另一种方法是根据速度设定点重新参数化油门，并由经典控制器输出油门以尝试匹配设定点。总体而言，在简单模拟器上进行的实验（第 IV-A 节）表明，连续动作虽然有点难以学习，但可以提供更平滑的控制器。我们使用二维动作空间；范围 $[-1, 1]$ 内的转向角和以 km/h 为单位的设定速度。

c) *Reward function*: 奖励函数的设计可以接近监督学习——给定车道分类系统，可以根据最小化与车道中心的预测距离来设置学习车道跟随的奖励，这是[8]中采用的方法。这种方法的规模有限：系统只能与手工奖励背后的人类直觉一样好。我们不采取这种方法。相反，我们将奖励定义为前进速度，并在违反交通规则时终止事件 - 因此给定状态 $V(s_t)$ 的值对应于违规之前行驶的平均距离。可以识别的一个错误是代理可能会选择避免更困难的操作，例如在英国右转（在美国左转）。在未来的工作中可以利用命令条件奖励来避免这种情况。

B. Reinforcement Learning Algorithm – Deep Deterministic Policy Gradients

我们选择了一种简单的连续动作域无模型强化学习算法：深度确定性策略梯度（DDPG）[8]，以表明一种现成的强化学习算法没有特定于任务的适应能力，能够解决第 III-A 节中提出的 MDP。

DDPG 由两个函数逼近器组成：一个批评者 $Q: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ ，它估计在状态 s 下使用动作 a 时的预期累积折扣奖励的值 $Q(s, a)$ ，经过训练以满足贝尔曼方程

$$Q(s_t, a_t) = r_{t+1} + \gamma(1 - d_t)Q(s_{t+1}, \pi(s_{t+1})),$$

根据参与者 $\pi: \mathcal{S} \rightarrow \mathcal{A}$ 给出的策略，尝试估计 Q 最优策略 $\pi(s) = \operatorname{argmax}_a Q(s, a)$ ；这里 $(s_t, a_t, r_{t+1}, d_{t+1}, s_{t+1})$ 是一个经验元组，是使用动作 a_t 从状态 s_t 到 s_{t+1} 的转换，并接收奖励 r_{t+1} 和“完成”标志 d_{t+1} ，从过去的经验缓冲区中选择。批评者试图最小化贝尔曼等式中的错误，称为时间差异 (TD) 错误。存在许多演员批评家方法的变体，参见例如[28]，[29]。

DDPG 培训在线完成。除了建立在真实车辆上使用的缓冲区的基础设施（这要求它能够容忍丢失/故障事件并且随时可停止）之外，还可以通过从重放缓冲区中选择最“信息丰富”的示例来加速强化学习。我们使用一种普遍建立的方法来做这一点，称为优先经验重播[30]：我们以与评论家所犯的 TD 错误成比例的概率对经验元组进行采样。用于此采样的权重会在每个优化步骤中以最小的开销进行更新；新样本被赋予无限权重，以确保所有样本至少被看到一次。

DDPG 是一种离策略学习算法，这意味着训练期间执行的操作来自与参与者学习最优策略不同的策略。发生这种情况是为了获得最优策略所看到的狭窄分布之外的多样化状态行为数据，从而提高稳健性。我们在连续强化学习方法的背景下使用标准方法来实现这一目标：我们的探索策略是通过将离散的 Ornstein-Uhlenbeck 过程噪声 [31] 添加到最优策略中形成的。因此，在每一步中，我们都会添加由下式给出的最优动作噪声 x_t ：

$$x_{t+1} = x_t + \theta(\mu - x_t) + \sigma \epsilon_t, \quad (4)$$

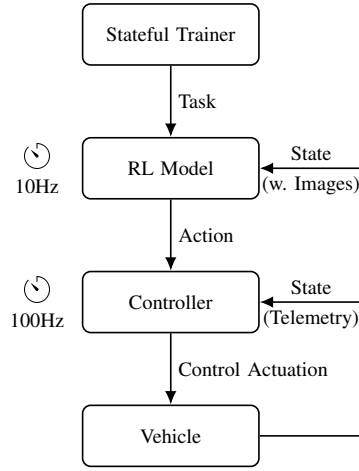
其中 θ, μ, σ 是超参数， $\{\epsilon_t\}_t$ 是独立同分布。从正态分布 $N(0, 1)$ 中采样的随机变量。这些参数需要仔细调整，因为噪声效用和安全驾驶员的舒适度之间存在直接权衡。具有较低方差的强均值恢复噪声更容易预测，而较高方差的噪声则提供更好的状态动作空间覆盖。


```

1: while True do
2:   Request task
3:   Waiting for environment reset
4:   if task is train then
5:     Run episode with noisy policy
6:     if exploration time is over then
7:       Optimise model
8:     end if
9:   else if task is test then
10:    Run episode with optimal policy
11:   else if task is undo then
12:    Revert previous train/test task
13:   else if task is done then
14:    Exit experiment
15:   end if
16: end while

```

(a) Task-based workflow for on-vehicle training



(b) Policy execution architecture, used to run episodes during model training or testing.

Fig. 2: Outline of the workflow and the architecture for efficiently training the algorithm from a safety driver’s feedback.

C. Task-based Training Architecture

Deployment of a reinforcement learning algorithm on a full-sized robotic vehicle running in a real world environment requires adjustment of common training procedures, to account for both driver intervention and external variables affecting the training.

We structure the architecture of the algorithm as a simple state machine, outlined in Figure 2a, in which the safety driver is in control of the different tasks. We define four tasks: train, test, undo and done. The definition of these tasks allows the system to be both interactive and stateful, favouring an on-demand execution of episodes instead of an a priori fixed schedule.

The train and test tasks allow us to interact with the vehicle in autonomous mode, executing the current policy. The difference between the two tasks consists in noise being added to the model output and the model being optimised in training tasks, whereas test tasks run directly the model output actions. During early episodes, we skip optimisation to favour exploration of the state space. We continue the experiment until the test reward stops increasing.

Each episode is executed until the system detects that automation is lost (i.e. the driver intervened). In a real world environment, the system can not reset automatically between episodes, unlike agents in simulation or in a constrained environment. We require a human driver to reset the vehicle to a valid starting state. Upon episode termination, while the safety driver performs this reset, the model is being optimised, minimising the time spent between episodes.

The undo and done tasks depict the key differences in the architecture. The system may terminate an episode for a variety of valid reasons other than failing to drive correctly: these episodes can not be considered for the purposes of training. The undo task is introduced for this reason, as it allows us to undo the episode and restore the model as it was before running that episode. A common example in our

experiments is encountering other drivers seeking to use the road being used as the environment. The done task allows us to gracefully exit the experiment at any given moment, and is helpful since the procedure is interactive and it doesn’t run for a fixed number of episodes.

IV. EXPERIMENTS

The main task we use to showcase the vehicle is that of lane-following; this is the same task as addressed in [8], however done on a real vehicle as well as on simulation, and done from image input, without knowledge of lane position. It is a task core to driving, and was the cornerstone of the seminal ALVINN [19]. We first accomplish this task in simulation in Section IV-A, and then use these results and knowledge of appropriate hyperparameters to demonstrate a solution on a real vehicle in Section IV-B.

For both simulation and real-world experiments we use a small convolutional neural network. Our model has four convolutional layers, with 3×3 kernels, stride of 2 and 16 feature dimensions, shared between the actor and critic models. We then flatten the encoded state and concatenate the vector the scalar state for the actor, additionally concatenating the actions for the critic network. For both networks we then apply one fully-connected layer with feature size 8 before regressing to the output. For the VAE experiments, a decoder of the same size as the encoder is used, replacing strided convolution with transposed convolution to upsample the features. A graphical depiction is shown in Figure 1.

A. Simulation

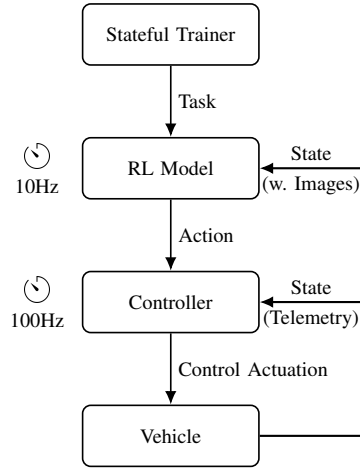
To test reinforcement learning algorithms in the context of lane following from image inputs we developed a 3D driving simulator, using Unreal Engine 4. It contains a generative model for country roads, supports varied weather conditions and road textures, and will in the future support more complex environments (see Figure 3 for game screenshots).

```

1: while True do
2:   Request task
3:   Waiting for environment reset
4:   if task is train then
5:     Run episode with noisy policy
6:     if exploration time is over then
7:       Optimise model
8:     end if
9:   else if task is test then
10:    Run episode with optimal policy
11:   else if task is undo then
12:    Revert previous train/test task
13:   else if task is done then
14:    Exit experiment
15:   end if
16: end while

```

(a) Task-based workflow for on-vehicle training



(b) Policy execution architecture, used to run episodes d模型 training or testing.

图 2: 根据安全驾驶员的反馈有效训练算法的工作流程和架构概述。

C. Task-based Training Architecture

在现实环境中运行的全尺寸机器人车辆上部署强化学习算法需要调整常见的训练程序，以考虑驾驶员干预和影响训练的外部变量。

我们将算法的架构构建为一个简单的状态机，如图 2a 所示，其中安全驱动程序控制不同的任务。我们定义了四个任务：训练、测试、撤消和完成。这些任务的定义允许系统既是交互式的又是有状态的，有利于按需执行情节而不是先验的固定时间表。

训练和测试任务使我们能够在自主模式下与车辆交互，执行当前的策略。这两个任务之间的区别在于，噪声被添加到模型输出中，并且模型在训练任务中被优化，而测试任务则直接运行模型输出操作。在早期阶段，我们跳过优化以支持对状态空间的探索。我们继续实验，直到测试奖励停止增加。

每个情节都会执行，直到系统检测到自动化丢失（即驾驶员干预）。在现实环境中，系统无法在情节之间自动重置，这与模拟或受限环境中的代理不同。我们需要人类驾驶员将车辆重置为有效的启动状态。事件终止后，当安全驱动程序执行此重置时，模型将得到优化，从而最大限度地减少事件之间花费的时间。

撤消和完成的任务描述了架构中的主要差异。除了无法正确驾驶之外，系统可能会出于各种有效原因终止事件：这些事件不能被视为训练目的。由于这个原因，引入了撤消任务，因为它允许我们撤消该情节并将模型恢复为运行该情节之前的状态。我们的一个常见的例子

实验是遇到其他驾驶员试图使用正在使用的道路作为环境。完成的任务允许我们在任何给定时刻优雅地退出实验，并且很有帮助，因为该过程是交互式的，并且不会运行固定数量的剧集。

四. 实验

我们用来展示车辆的主要任务是车道跟随；这与[8]中解决的任务相同，但是在真实车辆和模拟上完成的，并且是通过图像输入完成的，而不知道车道位置。它是驾驶的核心任务，也是开创性的 ALVINN [19] 的基石。我们首先在第 IV-A 节中的模拟中完成此任务，然后在第 IV-B 节中使用这些结果和适当超参数的知识在真实车辆上演示解决方案。

对于模拟和现实实验，我们使用小型卷积神经网络。我们的模型有四个卷积层，有 3 个 3×3 个内核，步长为 2 和 16 个特征维度，在演员模型和评论家模型之间共享。然后，我们展平编码状态并将向量与参与者的标量状态连接起来，另外还连接评论家网络的动作。然后，对于这两个网络，我们应用一个特征大小为 8 的全连接层，然后再回归到输出。对于 VAE 实验，使用与编码器大小相同的解码器，用转置卷积代替跨步卷积来对特征进行上采样。图形描述如图 1 所示。

A. Simulation

为了在图像输入的车道跟踪背景下测试强化学习算法，我们使用虚幻引擎 4 开发了一个 3D 驾驶模拟器。它包含乡村道路的生成模型，支持不同的天气条件和道路纹理，并且将来将支持更复杂的环境（参见图 3 的游戏屏幕截图）。

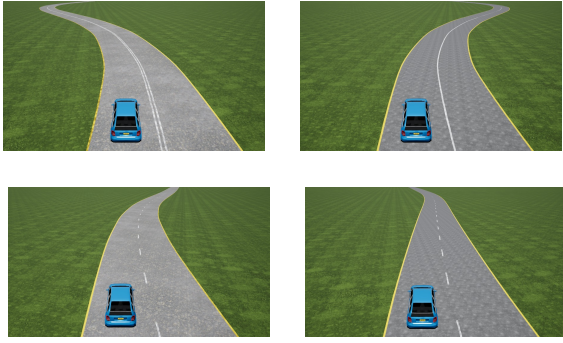


Fig. 3: Examples of different road environments randomly generated for each episode in our lane following simulator. We use procedural generation to randomly vary road texture, lane markings and road topology each episode. We train using a forward facing driver-view image as input.

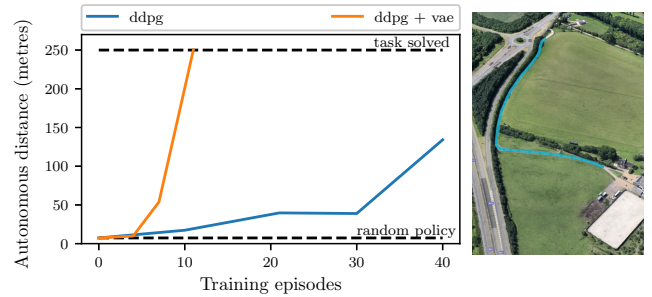
The simulator proved essential for tuning reinforcement learning parameters including: learning rates, number of gradient steps to take following each training episode and the correct termination procedure – conservative termination leads to a better policy. It confirmed a continuous action space is preferable – discrete led to a jerky policy – and that DDPG is a suitable reinforcement learning algorithm. As described in the environment setup in Section III-A, reward granted in the simulator corresponded to the distance travelled before exiting lane, with new episodes resetting the car to the centre of the lane.

We found that we could reliably learn to learn follow in simulation from raw images within 10 training episodes. Furthermore, we found little advantage to using a compressed state representation (provided by a Variational Autoencoder). We found the following hyperparameters to be most effective, which we use for our real world experiments: future discount factor of 0.9, noise half-life of 250 episodes, noise parameters of θ of 0.6 and σ of 0.4, 250 optimisation steps between episodes with batch size 64 and gradient clipping of 0.005.

B. Real-world driving

Our real world driving experiments mimic in many ways those conducted in simulation. However, executing this experiment in the real world is significantly more challenging. Many environmental factors cannot be controlled, and real-time safety and control systems must be implemented. For these experiments, we use a 250 meter section of road. The car begins at the start of the road to commence training episodes. When the car deviates from the lane and enters an unrecoverable position, the safety driver takes control of the vehicle ending the episode. The vehicle is then returned to the center of the lane to begin the next episode. We use the same hyperparameters that we found to be effective in simulation, with the noise model adjusted to give vehicle behaviour similar to that in simulation under the dynamics of the vehicle itself.

We conduct our experiments using a modified Renault



(a) Algorithm results

(b) Route

Fig. 4: Using a VAE with DDPG greatly improves data efficiency in training over DDPG from raw pixels, suggesting that state representation is an important consideration for applying reinforcement learning on real systems. The 250m driving route used for our experiments is shown on the right.

Twizy vehicle, which is a two seater electric vehicle, shown in Figure 1. The vehicle weighs 500kg, has a top speed of 80 km/h and has a range of 100km on a single battery charge. We use a single monocular forward-facing video camera mounted in the centre of the roof at the front of the vehicle. We use retrofitted electric motors to actuate the brake and steering, and electronically emulate the throttle position to regulate torque to the wheels. All computation is done on-board using a single NVIDIA Drive PX2 computer. The vehicle’s drive-by-wire automation automatically disengages if the safety driver intervenes, either by using vehicle controls (brake, throttle, or steering), toggling the automation mode, or pressing the emergency stop. An episode would terminate when either speed exceeded 10km/h, or drive-by-wire automation disengaged, indicating the safety driver has intervened. The safety driver would then reset the car to the centre of the road and continue with the next episode.

Table I shows the results of these experiments. Here, the major finding is that reinforcement learning can solve this problem in a handful of trials. Using 250 optimisation steps with batch size 64 took approximately 25 seconds, which made the experiment extremely manageable, considering manoeuvring the car to the centre of the lane to commence the next episode takes approximately 10 seconds anyway. We also observe in the real world, where the visual complexity is much more difficult than simulation, a compressed state representation provided by a Variational Autoencoder trained online together with the policy greatly improved reliability of the algorithm. We compare our method to a zero policy (driving straight with constant speed) and random exploration noise, in order to confirm that the trial indeed required a non-trivial policy.¹

V. DISCUSSION

This work presents the first application of deep reinforcement learning to a full sized autonomous vehicle. The

¹A video of the training process for our vehicle learning to drive the 250m length of private road with the stateful RL training architecture (Section III-C) is available at <https://wayve.ai/blog/l2diad>

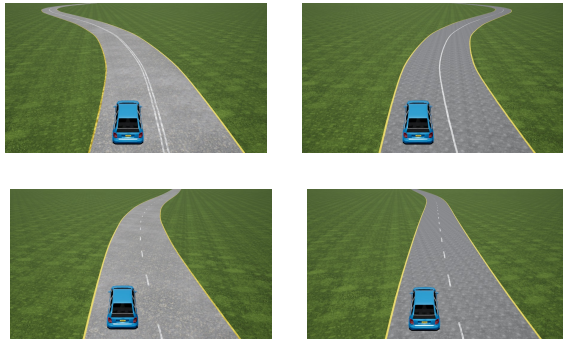


图 3: 在我们的车道跟随模拟器中为每个情节随机生成的不同道路环境的示例。我们使用程序生成来随机改变每集的道路纹理、车道标记和道路拓扑。我们使用前向驾驶员视图图像作为输入进行训练。

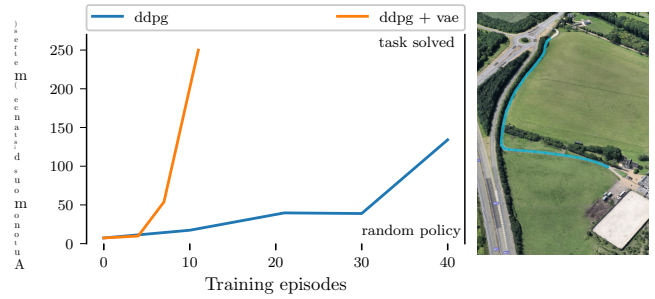
事实证明，模拟器对于调整强化学习参数至关重要，包括：学习率、每次训练后要采取的梯度步数以及正确的终止程序——保守的终止会带来更好的策略。它证实了连续的动作空间是更好的——离散的会导致策略不稳定——并且 DDPG 是一种合适的强化学习算法。正如第 II-A 节中的环境设置中所述，模拟器中授予的奖励与退出车道之前行驶的距离相对应，新的情节会将汽车重置到车道中心。

我们发现我们可以在 10 个训练集中可靠地从原始图像中学习模拟跟随。此外，我们发现使用压缩状态表示（由变分自动编码器提供）没有什么优势。我们发现以下超参数是最有效的，我们将其用于现实世界的实验：未来折扣因子为 0.9，噪声半衰期为 250 个片段，噪声参数 θ 为 0.6， σ 为 0.4，片段大小为 64，梯度裁剪为 0.005，片段之间有 250 个优化步骤。

B. Real-world driving

我们的现实世界驾驶实验在很多方面都模仿了模拟中进行的实验。然而，在现实世界中进行这个实验显然更具挑战性。许多环境因素无法控制，必须实施实时安全和控制系统。对于这些实验，我们使用一段 250 米的道路。汽车从道路起点开始开始训练。当汽车偏离车道并进入无法恢复的位置时，安全驾驶员控制车辆结束事件。然后车辆返回车道中心开始下一集。我们使用我们现在模拟中有效的相同超参数，调整噪声模型以使车辆行为与车辆本身动态下的模拟相似。

我们使用改装的雷诺进行实验



(a) Algorithm results

(b) Route

图 4: 使用带有 DDPG 的 VAE 极大地提高了从原始像素训练 DDPG 的数据效率，这表明状态表示是在实际系统上应用强化学习的一个重要考虑因素。我们实验中使用的 250m 行驶路线如右图所示。

Twizy 汽车是一款两座电动汽车，如图 1 所示。该车重 500 公斤，最高时速 80 公里/小时，一次充电续航里程为 100 公里。我们使用安装在车辆前部车顶中央的单目前向摄像机。我们使用改装的电动机来驱动制动和转向，并以电子方式模拟油门位置来调节车轮的扭矩。所有计算均使用一台 NVIDIA Drive PX2 计算机在机上完成。如果安全驾驶员通过使用车辆控制装置（制动、油门或转向）、切换自动化模式或按下紧急停止按钮进行干预，车辆的线控驾驶自动化系统将自动脱离。当速度超过 10 公里/小时或线控自动驾驶系统脱离时，事件就会终止，表明安全驾驶员已进行干预。然后安全驾驶员会将汽车重置到道路中央并继续下一集。

表一显示了这些实验的结果。在这里，主要发现是强化学习可以通过少量试验解决这个问题。使用批量大小为 64 的 250 个优化步骤大约需要 25 秒，这使得实验非常易于管理，考虑到将汽车操纵到车道中心以开始下一阶段大约需要 10 秒。我们还在现实世界中观察到，视觉复杂性比模拟困难得多，在线训练的变分自编码器提供的压缩状态表示与策略一起大大提高了算法的可靠性。我们将我们的方法与零策略（匀速直线行驶）和随机探索噪声进行比较，以确认试验确实需要一个不平凡的策略。¹

五、讨论

这项工作首次将深度强化学习应用于全尺寸自动驾驶车辆。这

¹ A video of the training process for our vehicle learning to drive the 250m length of private road with the stateful RL training architecture (Section III-C) is available at <https://wayve.ai/blog/l2diad>

Model	Training			Test	
	Episodes	Distance	Time	Meters per Disengagement	# Disengagements
Random Policy	-	-	-	7.35	34
Zero Policy	-	-	-	22.7	11
Deep RL from Pixels	35	298.8 m	37 min	143.2	1
Deep RL from VAE	11	195.5 m	15 min	-	0

TABLE I: Deep reinforcement learning results on an autonomous vehicle over a 250m length of road. We report the best performance for each model. We observe the baseline RL agent can learn to lane follow from scratch, while the VAE variant is much more efficient, learning to successfully drive the route after only 11 training episodes.

experiments demonstrate we are able to learn to lane follow with under thirty minutes of training – all done on on-board computers.

In order to tune hyperparameters, we built a simple simulated driving environment where we experimented with reinforcement learning algorithms, maximising distance before a traffic infraction using DDPG as a canonical algorithm. The parameters found transferred amicably to the real-world, where we rapidly trained a policy to drive a real vehicle on a private road, with a reward signal consisting only of speed and termination upon control driver taking control. Notably, this reward requires no further information or maps of the environment. With more data, vehicles and larger models, this framework is general enough to scale to more complex driving tasks.

Whilst viable, this approach will require translation of reinforcement learning research advancements, as well as work on core reinforcement learning algorithms if it is to become a leading approach for scaling autonomous driving. We conclude by discussing our thoughts on the future work required.

In this work, we present a general reward function which asks the agent to maximise the distance travelled without intervention from a safety driver. While this reward function is general, it has a number of limitations. It does not consider conditioning on a given navigation goal. Furthermore, it is incredibly sparse. As our agent improves, interventions will become significantly less frequent, resulting in weaker training signal. It is likely that further work is required to design a more effective reward function to learn a super-human driving agent. This will involve the careful consideration of many safety [32] and ethical issues [33].

The second area for development suggested by the results here is a better state representation. Our experiments have shown that a simple Variational Autoencoder greatly improves the performance of DDPG in the context of driving a real vehicle. Beyond pixel-space autoencoders is a wealth of computer vision research addressing effective compression of images: here existing work in areas such as semantic segmentation, depth, egomotion and pixel-flow provide an excellent prior for what is important in driving scenes [34], [1], [35]. This research needs to be integrated with reinforcement learning approaches for real tasks, both model-free and model-based.

However, unsupervised state encoding alone will likely not be sufficient. In order to compress the state in a manner that makes it simple to learn a policy with just a small number of

samples, information on which elements of the state (image observation) are important is required. This information should come from the reward and terminal signals. Reward and terminal information can be incorporated in an encoding in many ways, but one difficulty always prevails: credit allocation. Rewards obtained at a specific time step may be related to observations received many time-steps in the past. Thus good models used for this application will contain a temporal component.

Two areas that could greatly improve the availability of data for the application of reinforcement learning to real autonomous driving are semi-supervised learning [35] and domain transfer [36]. Whilst only a small portion of driving data might have rewards and terminals associated with it, as those are costly to obtain, the image embeddings – and perhaps other aspects of models – could benefit from driving data captured from dashcams in every-day vehicles. These could be used to pre-train the image autoencoder. In the context of a model-based RL system, these could also be used to approximate state transition functions, whilst advances in semi-supervised learning might allow us to utilise this data without reward/terminal labels data. Domain transfer, on the other hand, may allow us to create simulations sufficiently convincing that data from these may be used to train a policy that can be transferred directly onto a real car.

The algorithm used here is intentionally a common canonical approach, chosen to demonstrate the ease with which reinforcement learning may be applied to driving. Many improvements to it have been developed in the wider literature, including the use of natural gradients [37]. Other research has looked at better transformation of observations into states, typically using an RNN [38], [39], as well as methods to perform multi-step planning, as in [40]. It is no question that these could provide superior performance.

New advances in model-based reinforcement provide alternative exciting avenues for autonomous driving research, with work such as [22] showing outstanding performance of models when observing directly the state of a physical system. This could offer significant benefits to an image-based domain. Alternative model-based approaches include [41] which learn to simulate episodes and learn in imagination.

We hope this paper inspires more research into applying reinforcement learning research to autonomous driving, perhaps combining it with elements from other machine learning techniques such as imitation learning and control theory. The method here solved a simple driving task in half an hour – what more could be done in a day?

Model	Training			Test	
	Episodes	Distance	Time	Meters per Disengagement	# Disengagements
Random Policy	-	-	-	7.35	34
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Deep RL from Pixels	35	298.8 m	37 min	143.2	1
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表 I: 自动驾驶车辆在 250m 长的道路上进行深度强化学习的结果。我们报告每个模型的最佳性能。我们观察到基线 RL 智能体可以从头开始学习车道跟随，而 VAE 变体则更加高效，只需 11 个训练集即可学会成功驾驶路线。

实验表明，我们能够通过不到三十分钟的训练来学会车道跟随——所有这些都是车载计算机上完成的。

为了调整超参数，我们构建了一个简单的模拟驾驶环境，在其中尝试了强化学习算法，使用 DDPG 作为规范算法来最大化交通违规之前的距离。找到的参数被友好地转移到现实世界中，我们快速训练了一个在私人道路上驾驶真实车辆的策略，奖励信号仅包括速度和控制驾驶员控制后的终止。值得注意的是，这种奖励不需要更多的环境信息或地图。随着更多的数据、车辆和更大的模型，这个框架足够通用，可以扩展到更复杂的驾驶任务。

虽然可行，但如果要成为扩展自动驾驶的领先方法，这种方法将需要强化学习研究进展的转化，以及核心强化学习算法的研究。最后，我们讨论了我们对未来所需工作的想法。

在这项工作中，我们提出了一个通用奖励函数，要求智能体在没有安全驾驶员干预的情况下最大化行驶距离。虽然这种奖励函数很通用，但它有很多局限性。它不考虑对给定导航目标的调节。此外，它非常稀疏。随着我们的智能体的改进，干预的频率将显着降低，从而导致训练信号变弱。很可能需要进一步的工作来设计更有效的奖励函数来学习超人类驾驶代理。这将涉及许多安全[32]和道德问题[33]的仔细考虑。

这里的结果表明第二个发展领域是更好的国家代表性。我们的实验表明，简单的变分自动编码器极大地提高了 DDPG 在驾驶真实车辆情况下的性能。除了像素空间自动编码器之外，还有大量解决图像有效压缩问题的计算机视觉研究：语义分割、深度、自我运动和像素流等领域的现有工作为驾驶场景中的重要内容提供了良好的先验[34]、[1]、[35]。这项研究需要与实际任务的强化学习方法相结合，无论是无模型的还是基于模型的。

然而，仅无监督状态编码可能还不够。为了以某种方式压缩状态，使得仅用少量的学习策略变得简单

样本，需要有关状态（图像观察）的哪些元素很重要的信息。该信息应该来自奖励和终端信号。奖励和终端信息可以通过多种方式纳入编码中，但始终存在一个困难：信用分配。在特定时间步长获得的奖励可能与过去许多时间步长收到的观察结果有关。因此，用于此应用程序的良好模型将包含时间组件。

半监督学习[35]和领域迁移[36]可以极大地提高强化学习在实际自动驾驶中的应用数据的可用性。虽然只有一小部分驾驶数据可能有与之相关的奖励和终端，因为获取这些数据的成本很高，但图像嵌入——也许还有模型的其他方面——可以从日常车辆行车记录仪捕获的驾驶数据中受益。这些可用于预训练图像自动编码器。在基于模型的强化学习系统的背景下，这些也可以用于近似状态转换函数，而半监督学习的进步可能允许我们在没有奖励/终端标签数据的情况下利用这些数据。另一方面，域转移可以让我们创建足够令人信服的模拟，这些模拟中的数据可用于训练可以直接转移到真实汽车上的策略。

这里使用的算法有意成为一种常见的规范方法，选择它是为了证明强化学习可以轻松应用于驾驶。在更广泛的文献中已经对其进行了许多改进，包括自然梯度的使用[37]。其他研究着眼于将观察结果更好地转换为状态，通常使用 RNN [38]、[39]，以及执行多步骤规划的方法，如 [40] 中所示。毫无疑问，这些可以提供卓越的性能。

基于模型的强化的新进展为自动驾驶研究提供了令人兴奋的替代途径，[22]等工作显示了模型在直接观察物理系统状态时的出色性能。这可以为基于图像的领域带来显着的好处。基于模型的替代方法包括[41]，它学习模拟事件并在想象中学习。

我们希望本文能够激发更多将强化学习研究应用于自动驾驶的研究，或许将其与模仿学习和控制理论等其他机器学习技术的元素相结合。这里的方法在半小时内解决了一个简单的驾驶任务——一天还能完成什么？

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